COHOMOLOGY IS UNCHANGED BY PUSHING FORWARD ALONG A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma announces the aim of proving a theorem of Grothendieck and cites the original source there; that sentence lies outside the counted span and the cited bibliography entry is not needed for this card, whose counted spans carry no citation.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-cohomology-and-closed-immersions is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Pushforward along a closed immersion of topological spaces is exact. Modules, Lemma modules-lemma-i-star-exact, invoked at proof lines 3454-3455. Supplies exactness of the pushforward.
- Higher derived functors of an exact functor vanish. Derived Categories, Lemma derived-lemma-right-derived-exact-functor, invoked at proof lines 3456-3457. Supplies the vanishing of the higher derived functors of the pushforward.
- Leray spectral sequence for a composite of functors. Same chapter, Lemma lemma-apply-Leray, invoked at proof lines 3457-3458. Supplies the degeneration that yields the equality of cohomology groups.

Source reference. Original lemma 015F: sources/stacks-chow/tags.txt, source lines 1506–1506. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01F4: sources/stacks-cohomology/pinned-original-source.tex, source lines 2255–2255. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01AX: sources/stacks-chow/tags.txt, source lines 1706–1706. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of topological spaces. For any abelian sheaf $\mathcal{F}$ on Z we have $H^p(Z, \mathcal{F}) = H^p(X, i_*\mathcal{F})$.*

Proof. This is true because i_* is exact (see Modules, Lemma 01AX), and hence $R^p i_* = 0$ as a functor (Derived Categories, Lemma 015F). Thus we may apply Lemma 01F4. □